

Sneha Claire

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Summary

Machine Learning Engineer and Data Scientist with 5 years experience building scalable AI/ML systems and production-grade ML infrastructure, spanning supervised/unsupervised learning, computer vision, NLP, and real-time analytics. Strong foundations in Python, Docker, AWS, Linux, Git, and SQL.

Experience

Founding AI Engineer, Dogvatar – Cambridge, MA April 2025 – Present

- Built and deployed a scalable **RAG-LLM chatbot** from scratch at a service dog training startup, owning architecture, implementation, and productionization, integrating **FastAPI, Docker, and AWS Lambda, VPC, S3, Bedrock, RDS, and OpenSearch Serverless**.
- Co-defined the product roadmap and established a **CI/CD workflow** leveraging **AWS Amplify, Git, Docker, and CDK** to automate builds, testing, and deployments for the RAG-LLM API.
- Developed and customized a GPU-accelerated VLM inference pipeline on **AWS EKS** using **Terraform, Helm, and NVIDIA NIM**, enabling instant video-based training feedback and scalable multimodal model serving.
- Built a modular EC2 GPU-backed infrastructure with integrated monitoring, autoscaling, and persistent storage, designed to support both real-time VLM workloads and future fine-tuning of proprietary LLMs on puppy training datasets.
- Developed an automated **ETL** pipeline to ingest customer data from Salesforce into **S3** using **AWS Glue and EventBridge**, enabling scheduled, serverless data integration and pre-processing.
- Developing a scalable computer-vision ML framework for automated canine behavioral classification in training videos, reducing manual review load by 60–70%.

Machine Learning Engineer Co-op, Walmart – Andover, MA June 2024 – Dec 2024

- Developed a Python-based real-time dashboard using **Dash, Plotly, Docker** and **KQL** with integrated predictive modeling to deliver operational instructions to robotic system operators, boosting throughput and efficiency by over **25%**.
- Engineered end-to-end **ETL** pipelines to ingest and transform log data into **Azure Data Explorer**, enabling near-real-time operational tracking and feeding forecasting models for proactive decision-making.
- Built and validated predictive models with **Pandas, Scikit-Learn, and Statsmodels** to estimate robot stationary time during transit, optimizing scheduling and reducing idle periods.

Data Scientist II, Genpact – Bengaluru, KA, IN Oct 2020 – June 2023

- Led an 8-member team to develop a scalable ML framework using **Python, PyTorch, Pandas, scikit-learn, statsmodels, and Keras** for multiple large-scale analytics projects.
- Forecasted next-fiscal-year new customer counts for a global entertainment conglomerate's merchandise site by applying **SQL, Linear Regression, ARIMAX, Prophet, and LSTM**, achieving a **8%** prediction error.
- Built a subrogation vs. non-subrogation classifier for an insurance client with **98%** accuracy.
- Implemented data anonymization pipelines—masking, pseudo-randomization, and perturbation—using **fasttext, faker, and anonympy** to secure sensitive datasets.
- Built a provider–customer recommendation engine for an S&P 400 insurance company using **SQL, Python, Pandas, scikit-learn, and tensorflow recommenders** to streamline healthcare matchmaking.
- Applied K-prototypes and spectral-norm clustering to segment provider cohorts and designed a custom scoring algorithm to rank clusters for personalized recommendations.
- Developed an AWS NLP pipeline with **S3, Lambda Texttract, Elasticsearch, and Comprehend** to extract, index, and serve key information from PDF/DOC files, enabling search-driven provider suggestions.
- Developed an NLP classifier using **Python, TensorFlow, Elasticsearch, and spaCy** to map online search terms into franchises, departments, categories, and subdivisions, enhancing search-to-content relevance on a global entertainment merchandise site.

- Engineered a missed-sales-opportunity (MSO) detection pipeline: created an MSO dataset via DialogAct analysis and question clustering, and trained a **fastText** model to classify potential MSOs in chatbot interactions, driving proactive sales engagement.
- Applied unsupervised learning (k-means, DBSCAN, hierarchical clustering) with **SQL**, **pandas**, **scikit-learn**, and **statsmodels** in Python to infer customer age segments based on shopping-cart product profiles.
- Conducted profit and new-user signup analyses for a newly launched B2B service at a leading U.S. beverage company using **SQL**, **Python**, and **Pandas**, uncovering key performance drivers and growth opportunities.

Data Science Intern, Genpact – Bengaluru, KA, IN

July 2019 – Dec 2019

- Trained **HRNet** semantic segmentation framework for automated damage detection and classification on automotive imagery for a car insurance agency.
- Implemented Decision Tree and **LSTM** models in Python to forecast sales and cash-flow patterns for a leading U.S. fast-food chain, achieving a **12%** prediction error and identifying key business performance drivers.

Research Consultant, WorlQuant – Remote, IN

Oct 2018 – Sept 2019

- Developed quantitative alpha models to detect repeatable trading patterns in high volume market data.
- The quantitative models created are typically combined to make strategies used for portfolio management to maximize revenue.

Education

Boston University, MS in Computer Information Systems

Sept 2023 – Jan 2025

- **Coursework:** Artificial Intelligence, Data Structures and Algorithms, Data Science, Relational Database Management, Agile Software Development, IT Strategy and Management

BITS, Pilani, MS in Economics

Aug 2016 – May 2020

- **Coursework:** Econometrics, Statistics, Derivatives and Risk Management, Security Analysis and Portfolio Management, Business Analysis and Valuation

Projects

• **Research Fellow, Fatima Fellowship** – Remote, IN

May 2022 – Dec 2022

- Analyzed resilience of 20+ CNN architectures and activation layers using Python, PyTorch, Tensorflow, and Keras, applying error-injection techniques using PyTorchFI to assess resilience against soft errors.
- Researched adversarial robustness via stochastic weight sparsification, sampling per-layer Gaussian distributions $N_l(\alpha_l, \beta_l)$ and developing search algorithms to optimize sparsity parameters.
- Designed and implemented a GPU-accelerated neural network framework in C++ and CUDA, supporting extendable MLPs with optimized matrix multiplication, ReLU, and Sigmoid kernels.
- Developed neural network pruning pipelines in PyTorch, comparing local vs. global magnitude sparsity and prototyping custom pruning techniques to reduce model complexity.

Technologies

Languages: Python, SQL, C++, CUDA

Cloud & MLOps: AWS (Lambda, EKS, S3, IAM, Opensearch, EC2, EventBridge, Secrets Manager, VPC, CloudWatch, Cognito, Bedrock, Glue, RDS, Amplify), Docker, Terraform, Helm, CDK, CI/CD

ML & Data: Pandas, Langchain, Ragas, BeautifulSoup, PyTorch, TensorFlow, Keras, scikit-learn, Statsmodels, Plotly, Dash, Forecasting, Clustering, Recommender Systems

Services & Tools: FastAPI, ETL, Elasticsearch, Postgres, Git, Linux, Azure, GCP, Nvidia NIM, Jira